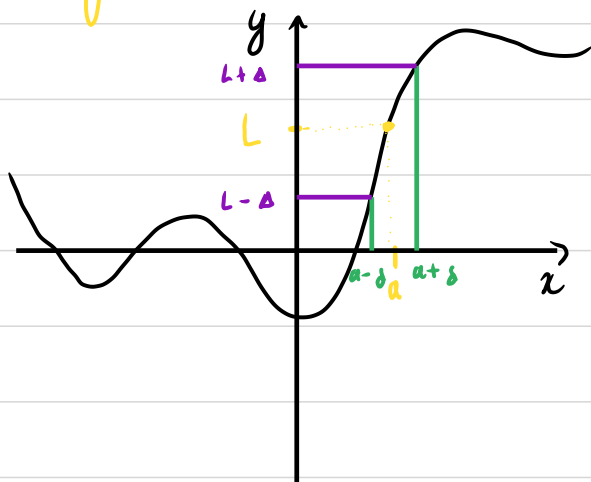


Limits

formal analysis



Suppose f is a real valued function defined for x arbitrarily close to a .

Definition: We say $f(x) \rightarrow L$ as $x \rightarrow a$ if for $\epsilon > 0$, there exists $\delta > 0$, such that for all x satisfying

$$0 < |x - a| < \delta, \text{ we have}$$

$$|f(x) - L| < \epsilon$$

We write $\lim_{x \rightarrow a} f(x) = L$

You don't actually need to define $f(x)$ at $x = a$ to determine the limit